# Meet EMMA: Your Emergency Medicine Mentoring Agent

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## Abstract

Medical licensing examinations demand synthesis of clinical reasoning across dozens of specialties under high-stakes conditions. This paper presents EMMA (Emergency Medicine Mentoring Agent), a conversational study agent that grounds responses in passages from eighteen standard medical textbooks via a five-component machine learning pipeline: biomedical NER for clinical entity extraction, FAISS-based dense retrieval over 36,723 textbook chunks, TF-IDF + LinearSVC specialty classification, BERTopic clustering for fine-grained topic routing, and a collaborative filtering recommender. A six-run ablation grid varying embedding model, LLM, and RAG condition shows that textbook-grounded RAG delivers up to an 11 percentage-point accuracy gain on MedQA under the right embedding–LLM pairing, while degrading performance under others. These findings suggest that RAG benefit is highly sensitive to semantic alignment between the embedding space and retrieval corpus.

***CCS Concepts:*** • **Computing methodologies → Natural language processing**; *Machine learning*.

***Keywords:*** retrieval-augmented generation, medical education, large language models, named entity recognition, topic modeling, collaborative filtering

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## 1 Introduction

Medical licensing examinations such as the United States Medical Licensing Examination (US-MLE) represent one of the most demanding knowledge challenges in professional education. Candidates must synthesize clinical reasoning across dozens of specialties under timed, high-stakes conditions.Traditional preparation relies on self-directed study, peer quizzing, and expensive tutoring — resources not equally accessible to all candidates.

This paper presents Emergency Medicine Mentoring Agent (EMMA), a conversational medical study agent built to close that gap. In explanation mode, students pose clinical questions in natural language and receive responses grounded in passages from eighteen standard medical textbooks. In quiz mode, EMMA presents authentic US-MLE-style questions, evaluates answers, and tracks per-specialty performance over time. A Collaborative Filtering (CF) recommender then steers students toward their weakest areas.

EMMA is built on a five-component Machine Learning (ML) pipeline: a SpaCy Named Entity Recognition (NER) model for clinical entity extraction, a Facebook AI Similarity Search (FAISS) vectorstore over 36,723 textbook chunks, a Term Frequency-Inverse Document Frequency (TF-IDF) + Linear Support Vector Classification (LinearSVC) specialty classifier, a BERTopic clustering model for fine-grained topic routing, and a Surprise-based CF recommender. All components feed a locally-run Qwen3-4B-Thinking-2507 Large Language Model (LLM). Figure 1 illustrates the full architecture.

The central empirical question is whether a textbook-grounded Retrieval-Augmented Generation (RAG) improves a small local LLM's accuracy on MedQA compared to a no-retrieval baseline. A six-run ablation grid varying embedding model, LLM variant, and RAG condition shows the answer is condition-dependent: with the right embedding and LLM pairing, RAG delivers a +11 percentage points (pp) accuracy gain; without those conditions, it can hurt. This nuanced finding is the primary contribution of this work.

The remainder is organized as follows: Section 2 formalizes the problem. Section 3 describes data preparation. Sections 4 through 7 present each pipeline component. Section 8 presents the ablation study, Section 9 evaluates results, Section 10 analyzes failure modes, and Section 11 concludes.

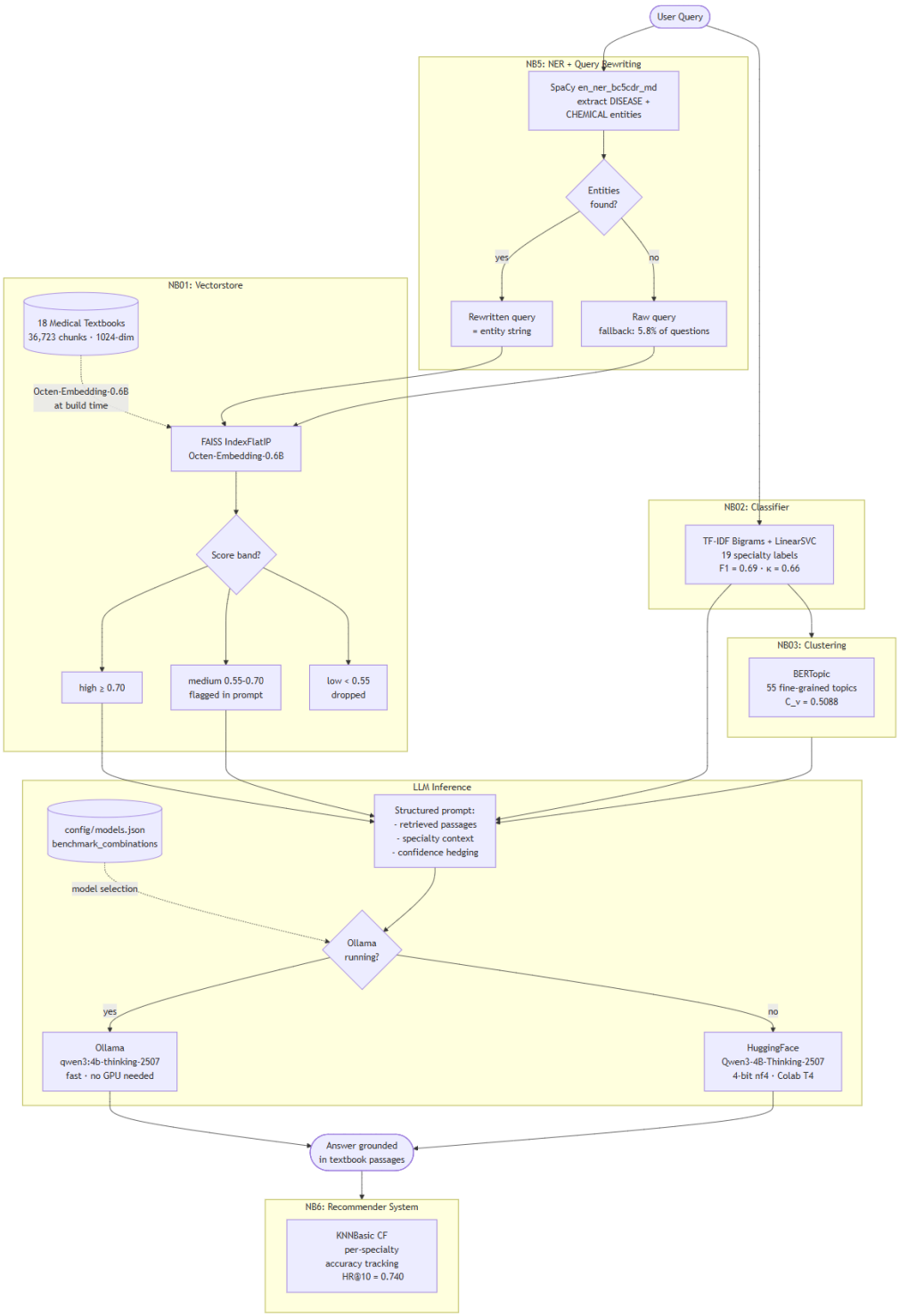


**Figure 1.** EMMA architecture pipeline. A student query passes through NER-based query rewriting, FAISS dense retrieval over 36,723 textbook chunks, specialty classification, BERTopic topic routing, and finally an LLM that generates a textbook-grounded response. The CF recommender operates in parallel on accumulated quiz performance.

## 2 Problem Formulation

Medical exam preparation requires extensive rote practice to internalise clinical vocabulary, symptom presentations, conditions, risk thresholds, and pharmacological details. Independent study is difficult without feedback, peer interaction, or access to professional tutoring — resources unevenly distributed across the candidate population. EMMA addresses this by providing an accessible, textbook-grounded conversational study agent capable of both answering clinical questions and administering adaptive quiz sessions.

This project builds directly on the Agentic Medical Graph-RAG (AMG-RAG) methodology of Rezaei et al. [12], using the same MedQA-US-MLE and MedMCQA datasets. Framing EMMA as a study agent rather than a triaging system reflects the structure of those datasets: their multiple-choice format, specialty labels, and associated textbook corpus are ideally suited to exam preparation rather than clinical decision support.

### 2.1 Vocabulary and Term Definitions

Before any ML component is applied, the input space must be anchored to clinically valid terminology. Patient and student queries may use lay language ("chest hurts", "can't breathe") that must be mapped to accepted medical terms (e.g., chest pain, dyspnea).

We define the accepted vocabulary $\Sigma$ as the set of clinical entity spans recognizable by a biomedical NER model (Section 4). Only tokens belonging to $\Sigma$ are extracted and used for FAISS query rewriting; input text is otherwise passed to the LLM in full.

Vocabulary coverage is extended by a Pointwise Mutual Information (PMI)-derived thesaurus mapping lay terms to clinical co-occurrents, constructed from collocation analysis of the MedQA training corpus (Section 4). Only tokens belonging to $\Sigma$ are extracted and used for FAISS query rewriting. Input text is otherwise passed to the LLM in full, separating the retrieval query (entity-focused, precise) from the generation prompt (full context, natural language).

### 2.2 Research Question

The central research question is: does textbook-grounded RAG improve small, locally-run LLM accuracy on MedQA compared to a no-retrieval baseline, and under what conditions?

This question is addressed by a six-run ablation grid varying embedding model (all-MiniLM-L12-v2, Qwen3-Embedding-0.6B, Octen-Embedding-0.6B), LLM (Qwen3-4B, Qwen3-4B-Thinking-2507), and RAG condition (on/off), evaluated on 100 MedQA questions per run.

### 2.3 Architecture

EMMA processes each user query through a five-stage pipeline before returning a response:

1. `en_ner_bc5cdr_md` extracts `DISEASE` and `CHEMICAL` entities from the raw query, replacing the full vignette text as the FAISS retrieval query. If no entities are found (5.8% of MedQA questions), the raw query is used unchanged.
2. The rewritten query is embedded and searched against a FAISS Index Flat Inner Product (IndexFlatIP) over the textbook vectorstore. Retrieved chunks are assigned confidence bands: high ($\geq 0.70$), medium (0.55–0.70, flagged in prompt), and low ($< 0.55$, dropped).
3. The raw query is independently passed to the TF-IDF + LinearSVC classifier, which assigns one of 19 specialty labels. The BERTopic clustering model provides finer-grained topic context within specialties.
4. All retrieved passages, entity spans, specialty labels, and confidence flags are assembled into a structured prompt with hedging instructions for low-confidence sources.
5. Inference is attempted via Ollama with no Graphics Processing Unit (GPU) required. If unavailable, the system falls back to HuggingFace transformers with 4-bit NormalFloat (NF4) quantization on a Colab T4 GPU. The production LLM is Qwen3-4B-Thinking-2507, selected by ablation (Section 8).

The CF recommender operates as a separate offline module, tracking per-specialty quiz accuracy and generating study recommendations independently of the RAG pipeline.

### 2.4 Model Selection Rationale

**2.4.1 Embedding Models.** Model selection followed the MTEB Leaderboard, specifically the RTEB Healthcare benchmark.

1. Qwen3-Embedding-0.6B was selected since its 8B variant, Qwen3-Embedding-8B, ranks #6 overall, the highest among models under 14B parameters (score 67.03). The 0.6B variant was chosen as the Video Random Access Memory (VRAM)-feasible member of that family given the Colab T4 budget.
2. Octen-Embedding-0.6B was selected because its 8B variant ranks #7 on the same leaderboard across all model sizes (score 73.79). Despite being in the same size class, it outperforms Qwen3-Embedding-8B.

Importantly, Octen-8B is derived from Qwen3-Embedding, meaning the 0.6B members of both families are directly comparable since the Octen variant is a domain-refined descendant. This made the two a natural ablation pair: same architecture family, different training regimen.

**2.4.2 LLM Models.** Model selection followed the Medmarks.ai benchmark (last updated December 2025), filtered to models under ~5B parameters.

1. Qwen3-4B-Thinking-2507 ranked #33 with a mean win rate of 0.4918 and weighted mean of 0.4883, substantially above all other sub-5B models evaluated.
2. The standard Qwen3-4B was included as a controlled baseline to isolate the effect of chain-of-thought reasoning, holding architecture and parameter count constant.

## 3 Data Preparation

The datasets used are MedQA-US-MLE [5] and MedMCQA [10], both of which are also used for benchmark evaluation in the AMG-RAG methodology of Rezaei et al. [12]. MedQA is a multiple-choice question answering dataset derived from professional medical licensing examinations [5], and its questions were written in reference to a collection of 18 standard medical textbooks. MedMCQA is built from medical entrance examination data and covers both foundational and clinical knowledge across a broad range of medical specialties [10].

For the purposes of this system, the MedQA textbooks serve as the RAG retrieval corpus, while MedMCQA is used exclusively to train the specialty classifier, as it provides explicit subject labels that MedQA lacks. Initial exploratory analysis confirmed answer balance in MedQA and characterised the specialty distribution in MedMCQA, with Internal Medicine dominating training data. These counts were subsequently verified prior to vectorstore construction.

Following these exploratory checks, the MedQA textbooks are chunked into 400-word sections with a 50-word overlap, producing 36,723 chunks across all 18 books. To support the embedding model ablation (Section 8), three separate FAISS vectorstores were constructed — one each for all-MiniLM-L12-v2, Qwen3-Embedding-0.6B, and Octen-Embedding-0.6B. All three fit within the Colab T4 VRAM budget at 0.6B parameters.

## 4 Named Entity Recognition & Collocation

### 4.1 Problem Definition

Before retrieved textbook passages can improve LLM accuracy, the FAISS retrieval query must be precise. Raw clinical vignettes dilute the embedding withincidental language: patient age, presentation verbs, conjunctions. A query like "A 45-year-old man presents with chest pain, diaphoresis, and ST elevation in leads II, III, aVF" embeds very differently from "chest pain diaphoresis ST elevationmyocardial infarction." NER-based query rewriting isolates only the clinically meaningful tokens before querying FAISS.

### 4.2 Model Choice

The `en_ner_bc5cdr_md` model [9] is trained on the BC5CDR corpus [7] and produces two typed entity labels: `DISEASE` and `CHEMICAL`, that map directly to what US-MLE questions test. `en_core_sci_md` was rejected as it outputs a single generic `ENTITY` label unsuitable for typed retrieval rewriting.

### 4.3 Corpus Analysis

Running `en_ner_bc5cdr_md` over all 10,178 MedQA training questions produced 54,256 total entities (`DISEASE`: 39,575; `CHEMICAL`: 14,681), with a mean of 5.33 entities per question. Of these, 593 questions (5.8%) contain zero entities and fall back to raw query retrieval.

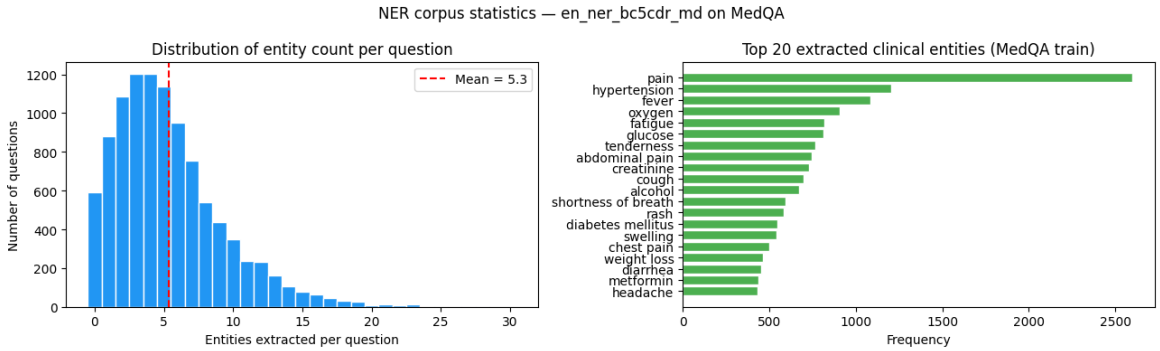


**Figure 2.** NER corpus statistics. Left: distribution of entity count per question (mean = 5.33; 5.8% of questions have zero entities). Right: top 20 extracted clinical entities by frequency — pain, hypertension, and fever dominate, reflecting the clinical reasoning focus of US-MLE questions.

### 4.4 Retrieval Score Impact

FAISS top-1 retrieval scores were compared for raw vignettes versus NER-rewritten entity strings across all three embedding models on a 50-question sample of clinical vignettes. Results: all-MiniLM-L12-v2 −0.022, Qwen3-Embedding-0.6B +0.005, Octen-Embedding-0.6B +0.006. NER rewriting hurts MiniLM because removing context reduces signal for a non-biomedical embedding.

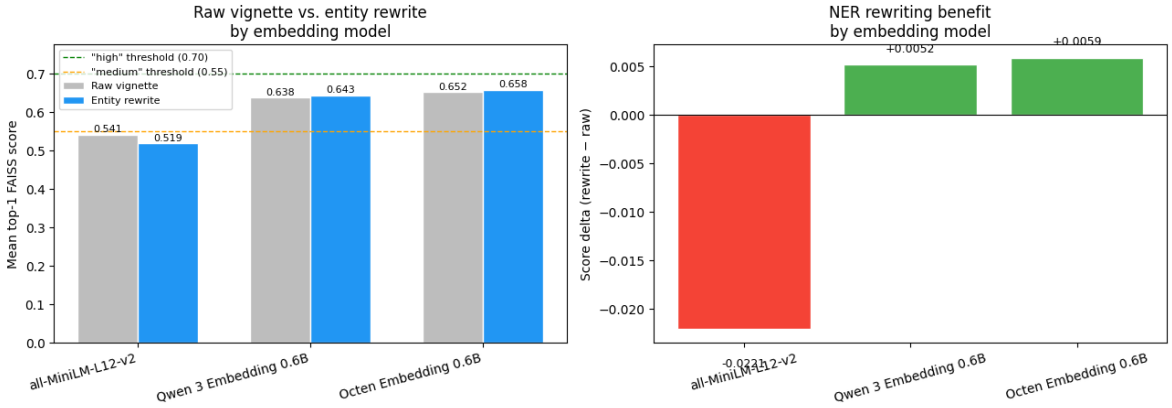


**Figure 3.** NER query rewriting impact on FAISS top-1 retrieval scores. Rewriting helps biomedical-scale embeddings and hurts the general-purpose MiniLM model, directly informing model selection for the ablation grid.

### 4.5 Collocation Analysis

PMI and chi-square bigram analysis was applied to all MedQA training question text (minimum frequency threshold: 10). Two hundred top bigrams were extracted by each measure. A PMI-derived lay-to-clinical thesaurus was constructed by filtering bigrams where one word appears in a curated lay-term seed list.

## 5 Classification

The classification component is formulated as a supervised 19-class text classification problem. Its purpose is not to produce the final answer, but to assign each medical question to a clinically meaningful specialty category for downstream routing. The classifier is trained on MedMCQA (which provides explicit `subject_name` labels) and applied to MedQA (which does not).

Formally, given an input question as a feature vector $x \in \mathbb{R}^d$, the classifier learns a function $F : x \rightarrow y$, where $y \in \{1, \ldots, 19\}$ denotes a specialty label. This routing narrows the FAISS search space to the most relevant medical domain.

### 5.1 Feature Engineering

The pipeline evaluates a feature-classifier grid across three representation spaces: Bag of Words (BOW), TF-IDF bigrams, and dense sentence embeddings from MiniLM-L12-v2. The mean inter-category TF-IDF cosine similarity on MedMCQA is 0.72, confirming the task is tractable for supervised classification while remaining non-trivial.

Among representations tested, TF-IDF with bigrams proved strongest. Local lexical patterns and short phrase combinations (e.g., "myocardial infarction", "gestational diabetes") are highly discriminative for specialty prediction — exactly the kind of signal that TF-IDF captures but that dense embeddings smooth over by emphasizing global semantic similarity.

### 5.2 Model Selection and Results

Model selection uses stratified 10-fold cross-validation on a 20k sample of MedMCQA, then retrains the champion on the full 179,777-question corpus. The champion configuration, TF-IDF Bigrams + LinearSVC [2], achieves:

- CV weighted F1: 0.5424 ± 0.0086; Cohen's $\kappa$: 0.5089 ± 0.0096
- Full-corpus holdout weighted F1: 0.69; Cohen's $\kappa$: 0.66

| | Configuration | Corpus | Weighted F1 | Cohen's kappa |
|---|---|---|---|---|
| 0 | TF-IDF Bigrams + LinearSVC | 19,990 (CV sample) | 0.5424 +/- 0.0086 | 0.5089 +/- 0.0096 |
| 1 | BOW + LinearSVC | 19,990 (CV sample) | 0.5042 +/- 0.0100 | 0.4683 +/- 0.0109 |
| 2 | MiniLM-L6-v2 (CPU proxy) + LinearSVC | 5,000 (sample) | 0.4635 +/- 0.0254 | 0.4297 +/- 0.0268 |
| 3 | BOW + MNB | 19,990 (CV sample) | 0.4590 +/- 0.0084 | 0.4294 +/- 0.0089 |
| 4 | TF-IDF Bigrams + MNB | 19,990 (CV sample) | 0.3581 +/- 0.0097 | 0.3378 +/- 0.0103 |
| 5 | Word2Vec + LinearSVC | 19,990 (CV sample) | 0.3573 +/- 0.0134 | 0.3288 +/- 0.0137 |

**Figure 4.** Weighted F1 and Cohen's $\kappa$ across all feature-classifier combinations. TF-IDF Bigrams + LinearSVC is the clear champion, outperforming all embedding-based and probabilistic baselines.

Competing baselines performed substantially worse: BOW + LinearSVC achieved F1 = 0.50, all-MiniLM-L12-v2 + LinearSVC achieved F1 = 0.46, and BOW + Multinomial Naïve Bayes (MNB) achieved F1 = 0.46. The full comparison is shown in Figure 4.

The superiority of linear margin-based models over probabilistic baselines is consistent with prior findings on medical text classification [11], suggesting it is not corpus-specific. For EMMA, the routing module must be computationally efficient, deterministic, and easy to retrain — LinearSVC satisfies all three requirements.

# 6 Clustering

The clustering component is formulated as an unsupervised topic discovery problem whose objective is to uncover latent structure in medical exam questions and provide fine-grained routing signals for retrieval. Unlike the classifier, which assigns each question to one of 19 fixed specialty labels, clustering operates without supervision and partitions the corpus into semantically coherent groups based on content.

Formally, given a set of documents $D = \{d_1, \ldots, d_n\}$, the clustering model learns a mapping $G : d \rightarrow C_k$, where $C_k$ denotes one of $K$ discovered clusters. In EMMA, these clusters are not final outputs but intermediate structures that refine retrieval by grouping together questions with similar clinical themes.

## 6.1 Methodology

The implementation uses BERTopic [3], which combines transformer-based embeddings with density-based clustering and topic representation. Each MedQA question is first embedded using MiniLM-L12-v2, then reduced in dimensionality using Uniform Manifold Approximation and Projection (UMAP) [8] and clustered with Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) [1] at `min_cluster_size=50` (approximately 0.4% of the corpus), keeping topics clinically meaningful without over-splitting. This allows the model to auto-determine $K$ rather than requiring it to be specified a priori.

BERTopic discovers 55 clusters, plus an outlier cluster (topic = $-1$) covering 34.7% of questions that do not form sufficiently dense groups. These outliers retain their specialty label and fall back to specialty-only routing.

## 6.2 Evaluation

Since clustering is unsupervised, specialty labels from the classifier serve as proxy ground truth for Cohen's $\kappa$ computation.

**Table 1.** Clustering model comparison. BERTopic's near-zero $\kappa$ reflects granularity mismatch with 19 specialty labels, not failure — $C_v = 0.51$ confirms internal topic coherence.

| Method | Cohen's $\kappa$ | Silhouette | $C_v$ |
|---|---|---|---|
| TF-IDF + LSA + GMM | 0.019 | — | — |
| Embeddings + Spectral | 0.019 | 0.061 | — |
| BERTopic | -0.012 | 0.069 | 0.509 |

The key insight is that topic-level structure is more granular than specialty-level categorization. BERTopic discovers 55 fine-grained topics that do not align one-to-one with 19 specialty labels — this is granularity mismatch, not failure. Topic 0 (chest/cardiac terms) is 70.6% Internal Medicine; Topic 4 (gestation/pregnancy) is 72.4% Obstetrics. Figure 5 shows specialty purity across all 55 topics.

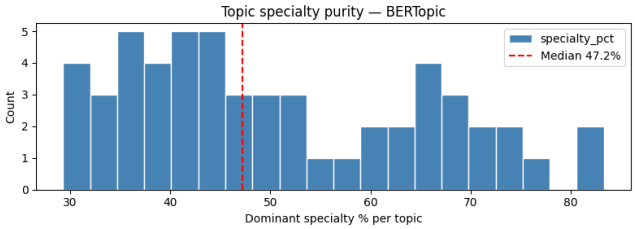


**Figure 5.** Topic specialty purity by BERTopic cluster. Several topics show high purity (>70%) for a single specialty, confirming clinically coherent grouping.

Together, classification provides coarse-grained specialty routing and clustering provides fine-grained topic refinement within that specialty. Questions assigned to the outlier cluster fall back gracefully to specialty-only routing.

# 7 Recommender System

The recommender component personalizes EMMA's quiz mode by directing students toward their weakest specialties based on accumulated session performance. CF is used because it identifies latent co-weakness patterns across students: a student weak in Anatomy is systematically more likely to be weak in Surgery, even before being tested in Surgery directly.

In the absence of real session data, a synthetic dataset of 200 students was constructed with Gaussian-distributed accuracy scores: each student was assigned 1–2 weak specialties (mean accuracy $\approx 0.30$) and stronger performance elsewhere (mean $\approx 0.75$), producing a score distribution centered around 77%. Four algorithms from the Surprise library [4] were evaluated using 5-fold Cross-Validation (CV) with Leave-One-Out (LOO) evaluation for ranking metrics:

1. Singular Value Decomposition (SVD): matrix factorization decomposing the student-specialty performance matrix into latent factors.
2. Non-negative Matrix Factorization (NMF): non-negative matrix factorization with non-negativity constraints.
3. K-Nearest Neighbors Basic (KNNBasic): neighbourhood-based CF identifying similar students by performance profile and recommending specialties similar students are weak in.
4. **NormalPredictor**: baseline predicting ratings from observed distribution, without considering student similarity or latent factors.

**Table 2.** Recommender algorithm comparison across all evaluation metrics. Precision@K (P@K) is structurally capped since students have only 3–4 weak specialties (theoretical max P@5 = 0.60). Hit Rate@K (HR@K) is the more informative metric — KNNBasic is the champion.

| Algorithm | RMSE | MAE | P@5 | HR@5 | P@10 | HR@10 |
|---|---|---|---|---|---|---|
| SVD | 0.2358 ± 0.0116 | 0.1880 ± 0.0092 | 0.067 | 0.335 | 0.054 | 0.540 |
| NMF | 0.2658 ± 0.0085 | 0.2224 ± 0.0087 | 0.051 | 0.255 | 0.060 | 0.595 |
| KNNBasic | 0.2208 ± 0.0105 | 0.1772 ± 0.0078 | 0.067 | 0.335 | 0.074 | **0.740** |
| NormalPredictor | 0.3109 ± 0.0116 | 0.2470 ± 0.0099 | 0.052 | 0.260 | 0.048 | 0.480 |

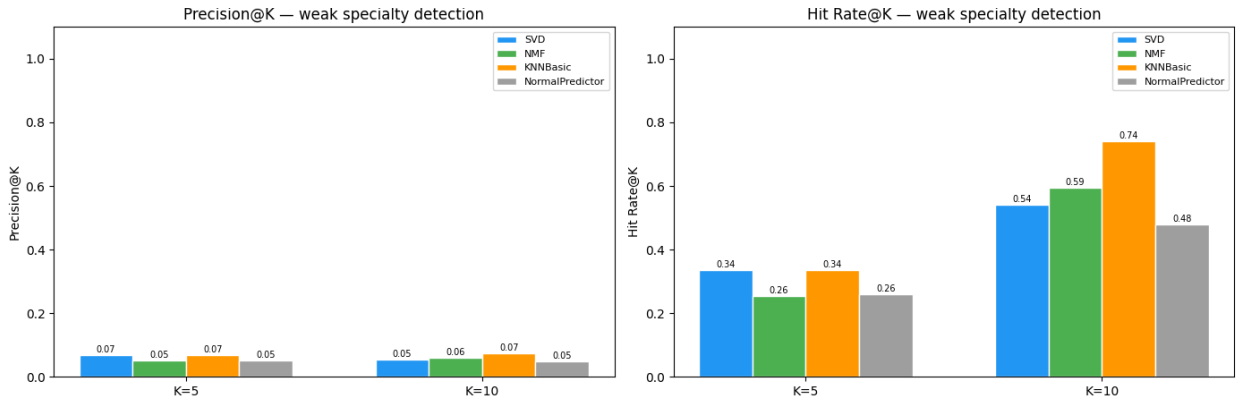


**Figure 6.** Hit Rate at K=5 and K=10 across algorithms. The consistent gap between HR@5 and HR@10 reflects the structural P@K ceiling — students have fewer weak specialties than K.

P@K scores are low across all algorithms (0.051–0.067 at K=5), but this is a ceiling effect rather than failure: students have only 3–4 weak specialties, making perfect Precision@5 impossible by construction. The theoretical maximum P@5 when a student has 3 weak specialties is 0.60. HR@K is the more informative metric.

The champion algorithm is KNNBasic, achieving the lowest Root Mean Square Error (RMSE) (0.2208 vs. NormalPredictor's 0.3109, a 29% reduction) and the highest Hit Rate at both K values. KNNBasic identifies at least one true weak specialty for 74% of students at K=10, compared to 48% for NormalPredictor. SVD achieves a comparable HR@10 of 54% but with a higher RMSE of 0.2358. KNNBasic's neighbourhood-based approach is particularly well suited to this setting: students with similar weakness profiles cluster tightly in the rating space, making nearest-neighbour more effective than the global latent structure recovered by matrix factorization.

# 8 RAG Pipeline & Ablation

## 8.1 Pilot Run

A pilot evaluation used Qwen3-4B (standard, non-thinking) with Qwen3-Embedding-0.6B on 100 MedQA questions. Retrieval used a FAISS IndexFlatIP [6] over the textbook vectorstore. Results: RAG accuracy = 38%, baseline = 42%, delta = −4 percentage points. RAG helped 6 questions and hurt 8. Mean retrieval score = 0.648. This result prompted the hypothesis that the standard LLM variant cannot effectively use retrieved context, motivating the full ablation.

## 8.2 Full Ablation Grid

A six-run grid was conducted varying embedding model, LLM variant, and RAG condition, with 100 MedQA questions per run. Table 3 summarizes the results. For reference, AMG-RAG [12] achieves 74.1% on MedQA using GPT-4o-mini (~8B parameters) with a dynamic PubMed knowledge graph.

**Table 3.** Six-run ablation grid. RAG helps only when both the embedding is biomedical-grade and the LLM can reason over retrieved passages.

| Run | Embedding | LLM | RAG | Acc. (%) |
|---|---|---|---|---|
| 1 | Qwen3-Emb | Qwen3-4B | off | 42 |
| 2 | Qwen3-Emb | Qwen3-4B | on | 38 |
| 3 | Qwen3-Emb | Thinking-2507 | off | 33 |
| 4 | Qwen3-Emb | Thinking-2507 | on | 34 |
| 5 | Octen-Emb | Thinking-2507 | off | 33 |
| 6 | Octen-Emb | Thinking-2507 | on | **44** |

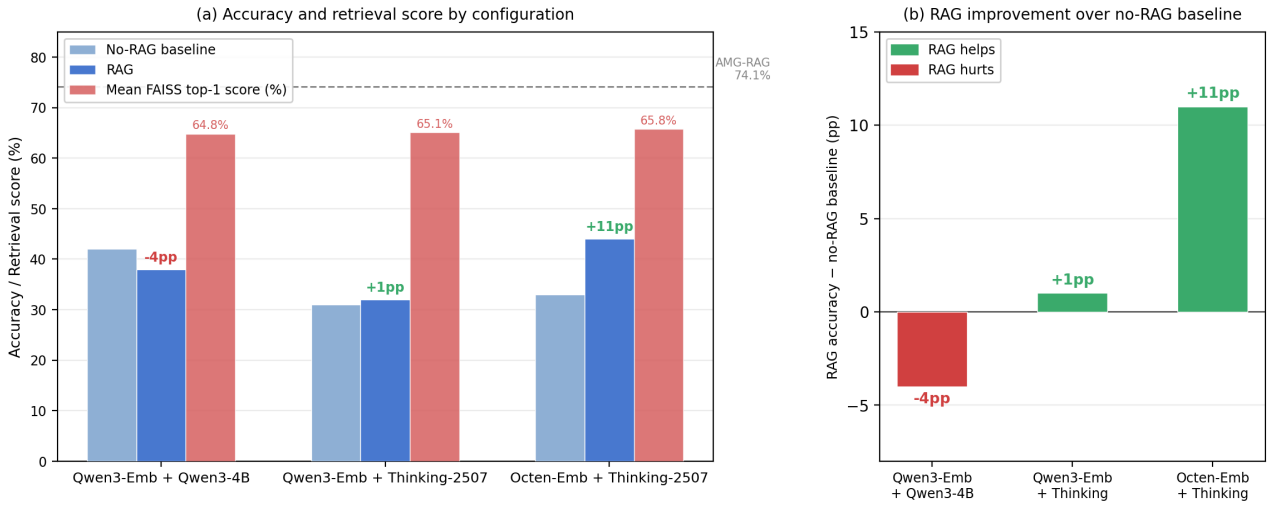


**Figure 7.** Ablation grid results. Run 6 (Octen-Embedding + Thinking-2507 + RAG) achieves 44% accuracy, a +11pp gain over its no-RAG baseline (run 5, 33%).

### 8.3 Interpretation

The ablation reveals three findings. First, standard Qwen3-4B cannot use retrieved context effectively: RAG hurts by −4pp regardless of embedding (runs 1–2). Second, switching to the Thinking variant alone is insufficient: with Qwen3-Embedding, RAG adds only +1 *pp* (runs 3–4). Third, the combination of a biomedical-grade embedding with a reasoning-capable LLM is required: Octen-Embedding + Thinking-2507 delivers +11 *pp* (runs 5–6).

This answers the research question affirmatively but conditionally: RAG improves accuracy only when the embedding is biomedical-grade *and* the LLM can reason over retrieved passages. The 30pp gap to AMG-RAG (74.1%) reflects LLM parameter scale (4B vs. ~8B), static vs. dynamic knowledge retrieval, and the absence of fine-tuning on the MedQA retrieval task specifically. Both systems evaluate on MedQA, so dataset difficulty is not a differentiating factor.

## 9 Evaluation

### 9.1 Classification Performance

The champion classifier achieves weighted F1 = 0.69 and Cohen's $\kappa$ = 0.66 on the held-out validation set, up from CV scores of 0.54 ± 0.01 and 0.51 ± 0.01 respectively — the improvement reflects retraining on the full 179,777-question corpus. The inter-category similarity of 0.71 confirms categories are moderately distinct — tractable for a linear classifier while remaining a non-trivial 19-class problem. The dominance of TF-IDF bigrams over dense embeddings follows directly: specialty labels are strongly signalled by domain-specific terminology that sparse frequency features capture more precisely than global semantic encodings.

### 9.2 Clustering Performance

Clustering evaluation confirms the granularity mismatch interpretation from Section 6: BERTopic's $C_v$ = 0.509 confirms internal coherence despite near-zero $\kappa$, and the 34.7% outlier rate reflects short question stems rather than model failure. From a systems perspective, clustering enhances retrieval precision even when assignments are imperfect — outlier questions fall back to specialty-only routing without penalty, ensuring no queries are lost.

### 9.3 NER as a Retrieval Quality Gate

The NER retrieval score results (Section 4) directly explain the embedding sensitivity observed in the ablation: the +0.006 Octen improvement vs. −0.022 MiniLM penalty translates to the +11pp vs. −4pp RAG delta in runs 5–6 vs. 1–2. The NER component is therefore necessary but not sufficient: its benefit is only realized when the downstream embedding can interpret biomedical entity strings. Embedding and NER strategy must be co-designed.

## 10 Error Analysis

### 10.1 Classification Errors

After selecting the champion model, a 90/10 holdout split was used to generate a per-specialty breakdown and confusion matrix. Dermatology (F1 = 0.30) is the most poorly classified specialty, consistent with its small training support (177 examples) and high lexical overlap with Pathology. Obstetrics & Gynaecology (F1 = 0.81) and Ophthalmology (F1 = 0.80) are most accurately classified, reflecting their distinctive specialized vocabularies.

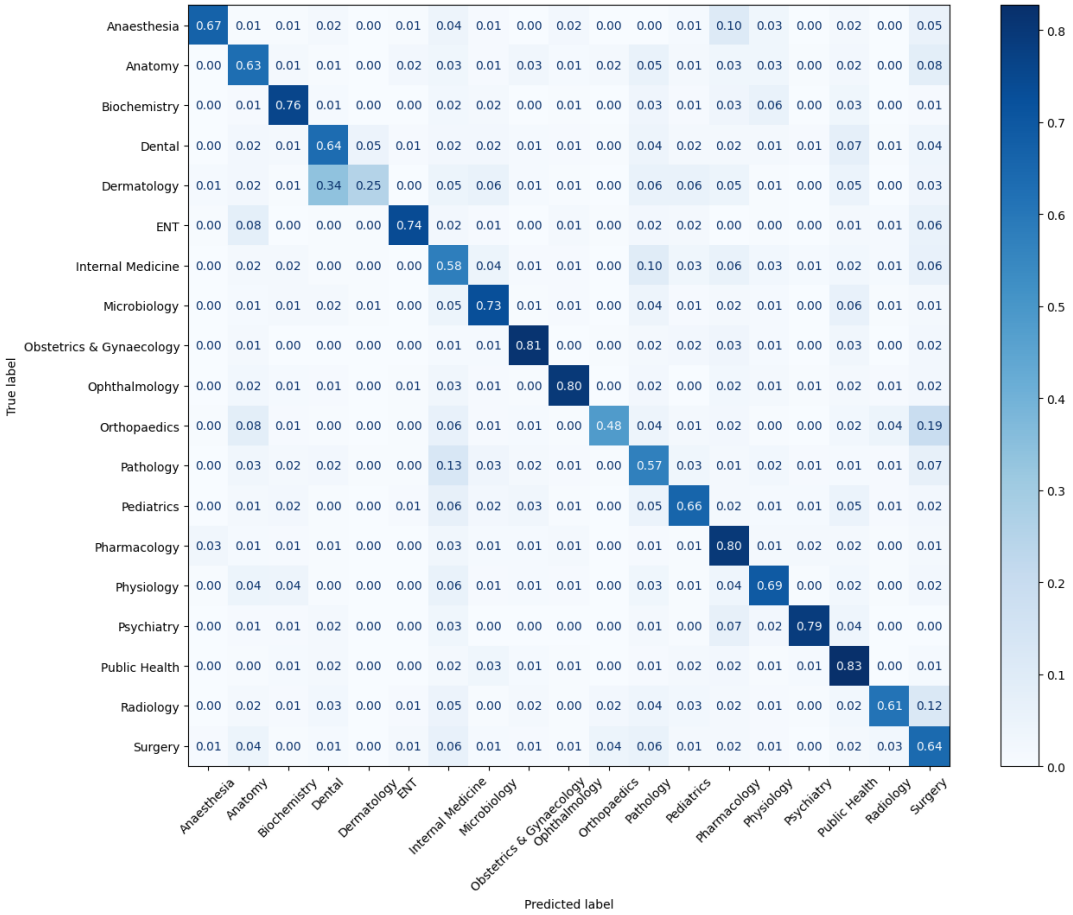


**Figure 8.** Champion classifier confusion matrix. Internal Medicine and Surgery show the most misclassification mass, consistent with their broad clinical overlap with other specialties.

```
Holdout weighted F1 : 0.6852
Holdout Cohen's kappa: 0.6639

                          precision    recall  f1-score   support

             Anaesthesia       0.74      0.67      0.70       317
                 Anatomy       0.71      0.63      0.67      1456
            Biochemistry       0.74      0.76      0.75       828
                  Dental       0.71      0.64      0.67       894
            Dermatology       0.37      0.25      0.30       177
                    ENT       0.80      0.74      0.77       492
       Internal Medicine       0.59      0.58      0.59      1789
            Microbiology       0.72      0.73      0.72      1131
Obstetrics & Gynaecology       0.80      0.81      0.81      1001
           Ophthalmology       0.80      0.80      0.80       693
            Orthopaedics       0.53      0.48      0.50       300
               Pathology       0.57      0.57      0.57      1489
              Pediatrics       0.65      0.66      0.66       804
            Pharmacology       0.69      0.80      0.74      1376
               Physiology       0.67      0.69      0.68       883
               Psychiatry       0.75      0.79      0.77       444
           Public Health       0.76      0.83      0.79      1778
               Radiology       0.66      0.61      0.63       440
                 Surgery       0.62      0.64      0.63      1686

                accuracy                           0.69     17978
               macro avg       0.68      0.67      0.67     17978
            weighted avg       0.69      0.69      0.69     17978
```

**Figure 9.** Per-specialty classification performance for the champion model. Dermatology (F1 = 0.30) is most challenging; Obstetrics & Gynaecology (F1 = 0.81) is most accurately classified.

### 10.2 RAG Pipeline Errors

Figure 10 shows RAG accuracy delta by specialty under the pilot configuration (Qwen3-4B + Qwen3-Embedding, run 2). RAG hurts Surgery (−40.0pp), Pathology (−14.2pp), Dental, and Public Health — specialties relying on clinical judgment and procedural reasoning rather than factual recall, where textbook retrieval is imprecise. Under the optimal configuration (run 6), Pharmacology and Obstetrics & Gynaecology show positive RAG deltas, consistent with their reliance on specific drug names and protocol thresholds.

### 10.3 NER Failure Modes

Of 10,178 MedQA training questions, 593 (5.8%) contain zero `DISEASE` or `CHEMICAL` entities and cannot benefit from NER rewriting. These questions typically involve mechanism-of-action or statistical reasoning ("What is the positive predictive value of…") rather than named conditions or drugs. They fall back to raw query retrieval without penalty.

| | specialty | n | RAG acc% | Base acc% | Delta |
|---|---|---|---|---|---|
| 0 | Internal Medicine | 14 | 28.6 | 28.6 | 0.0 |
| 1 | Pathology | 7 | 42.9 | 57.1 | -14.2 |
| 2 | Surgery | 5 | 20.0 | 60.0 | -40.0 |
| 3 | Microbiology | 4 | 50.0 | 25.0 | 25.0 |
| 4 | Psychiatry | 4 | 25.0 | 25.0 | 0.0 |
| 5 | Public Health | 3 | 33.3 | 66.7 | -33.4 |
| 6 | Pharmacology | 3 | 66.7 | 66.7 | 0.0 |
| 7 | Obstetrics & Gynaecology | 3 | 66.7 | 33.3 | 33.4 |
| 8 | Anatomy | 2 | 0.0 | 0.0 | 0.0 |
| 9 | Dental | 2 | 0.0 | 50.0 | -50.0 |
| 10 | Pediatrics | 2 | 100.0 | 50.0 | 50.0 |
| 11 | Orthopaedics | 1 | 100.0 | 100.0 | 0.0 |

**Figure 10.** RAG accuracy delta by specialty under the pilot configuration (Qwen3-4B + Qwen3-Embedding). Surgery (−40.0pp) and Pathology (−14.2pp) are most harmed; Pharmacology benefits most under the optimal configuration.

### 10.4 Recommender Limitations

The CF recommender is trained on a synthetic 200-student dataset. The P@K ceiling effect (Section 7) and the small dataset size are the two primary limitations. Real deployment would accumulate genuine session data, improving CF performance and eliminating the need for synthetic rating generation.

## 11 Conclusion

This paper demonstrates that textbook-grounded RAG can improve small local LLM accuracy on MedQA, but the improvement is conditional on two factors: the embedding model must be biomedical-grade, and the LLM must be capable of reasoning over retrieved passages. The six-run ablation grid establishes this clearly. Standard Qwen3-4B cannot use retrieved context regardless of embedding, resulting in a −4pp penalty. The Thinking variant paired with Octen-Embedding-0.6B achieves 44% accuracy with RAG, a +11pp gain over the 33% no-RAG baseline.

The 30pp gap to AMG-RAG (74.1%) reflects LLM parameter scale, static vs. dynamic retrieval, and the absence of task-specific fine-tuning(Section 8). Closing it would require scaling beyond the T4 VRAM budget and replacing the static textbook corpus with live PubMed querying.

The five pipeline components — NER (F1-improving for biomedical embeddings), classification (F1 = 0.69, 19 specialties), clustering (55 coherent topics), RAG (+11pp under optimal conditions), and KNNBasic recommender (HR@10 = 0.740) — constitute a functional, evaluated, and reproducible medical study agent.

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